

Assignment 1

Section A: Axioms, Set Theory, and Limits

Question 1

The event “infinitely many A_n occur” is

$$\limsup_n A_n = \bigcap_{n=1}^{\infty} \bigcup_{k=n}^{\infty} A_k,$$

since ω lies in infinitely many A_n iff for every n there is some $k \geq n$ with $\omega \in A_k$.

Let $B_n = \bigcup_{k=n}^{\infty} A_k$. Then $\limsup_n A_n \subseteq B_n$ for every n , so by monotonicity and Boole’s inequality,

$$P\left(\limsup_n A_n\right) \leq P(B_n) \leq \sum_{k=n}^{\infty} P(A_k).$$

Since $\sum_k P(A_k) < \infty$, its tail $\sum_{k=n}^{\infty} P(A_k) \rightarrow 0$. The left side is fixed and bounded by something tending to 0, so

$$P\left(\limsup_n A_n\right) = 0.$$

This is the first Borel-Cantelli lemma; no independence is needed.

Question 2

By De Morgan and Boole’s inequality,

$$P\left(\bigcap_{i=1}^n A_i\right) = 1 - P\left(\bigcup_{i=1}^n A_i^c\right) \geq 1 - \sum_{i=1}^n (1 - p_i) = \sum_{i=1}^n p_i - (n - 1).$$

Since probabilities are nonnegative, the bound is $\max(0, \sum_i p_i - (n - 1))$.

Equality holds when the failure events A_i^c are pairwise disjoint. For instance, take $\Omega = \{0, 1, \dots, n\}$ with $P(\{i\}) = 1 - p_i$ and $P(\{0\}) = \sum_i p_i - (n - 1)$, and set $A_i = \Omega \setminus \{i\}$, so $A_i^c = \{i\}$ and $P(A_i) = p_i$. Then $\bigcap_i A_i = \{0\}$ and

$$P\left(\bigcap_i A_i\right) = \sum_i p_i - (n - 1),$$

hitting the bound exactly.

Question 3

True. Multiplying $P(A | B) \geq P(A)$ by $P(B) > 0$ gives the symmetric form $P(A \cap B) \geq P(A)P(B)$. By inclusion-exclusion,

$$P(A^c \cap B^c) = 1 - P(A) - P(B) + P(A \cap B) \geq 1 - P(A) - P(B) + P(A)P(B) = (1 - P(A))(1 - P(B)) = P(A^c)P(B^c)$$

Dividing by $P(B^c)$ gives $P(A^c | B^c) \geq P(A^c)$.

(Note the crossed version $P(A | B^c) \geq P(A)$ is false, since it flips sign.)

Section B: Combinatorics & Classical Probability

Question 4

The number of tagged animals X in the sample is hypergeometric, so the probability of the observed outcome as a function of N is

$$L(N) = \frac{\binom{T}{k} \binom{N-T}{n-k}}{\binom{N}{n}}.$$

Since N is discrete, examine the ratio

$$\frac{L(N)}{L(N-1)} = \frac{\binom{N-T}{n-k} \binom{N-1}{n}}{\binom{N-1-T}{n-k} \binom{N}{n}} = \frac{(N-T)(N-n)}{N(N-T-n+k)}.$$

This ratio is ≥ 1 iff

$$(N-T)(N-n) \geq N(N-T-n+k) \iff -Tn \geq -TN \cdot \frac{k}{T} \iff Nk \leq nT,$$

that is $N \leq nT/k$. So L increases while $N \leq nT/k$ and decreases after, giving the maximizer

$$\hat{N} = \left\lfloor \frac{nT}{k} \right\rfloor.$$

Question 5

Exactly k correct means choosing the k fixed points and deranging the rest:

$$P_n(k) = \frac{\binom{n}{k} D_{n-k}}{n!}, \quad D_m = m! \sum_{j=0}^m \frac{(-1)^j}{j!}.$$

Expanding $\binom{n}{k} = \frac{n!}{k!(n-k)!}$ and $D_{n-k} = (n-k)! \sum_{j=0}^{n-k} \frac{(-1)^j}{j!}$, the factorials cancel:

$$P_n(k) = \frac{1}{k!} \sum_{j=0}^{n-k} \frac{(-1)^j}{j!}.$$

As $n \rightarrow \infty$ the sum is the partial series for e^{-1} , so

$$\lim_{n \rightarrow \infty} P_n(k) = \frac{1}{k!} \sum_{j=0}^{\infty} \frac{(-1)^j}{j!} = \frac{e^{-1}}{k!} = \frac{1}{e \cdot k!},$$

which is the Poisson(1) pmf.

Question 6

Each draw adds c balls of the opposite color, so the total grows by c each time. With $N_0 = r + b$,

$$P(R_1) = \frac{r}{r+b}.$$

For R_2 , condition on draw 1. If draw 1 is red (prob $\frac{r}{r+b}$), the urn becomes r red, $b+c$ blue; if blue (prob $\frac{b}{r+b}$), it becomes $r+c$ red, b blue. Total is $r+b+c$ either way:

$$P(R_2) = \frac{r}{r+b} \cdot \frac{r}{r+b+c} + \frac{b}{r+b} \cdot \frac{r+c}{r+b+c} = \frac{r(r+b) + bc}{(r+b)(r+b+c)}.$$

For R_3 , condition on the colors of draws 1 and 2 (four cases), with total $r+b+2c$ at draw 3:

$$P(R_3) = \frac{1}{(r+b)(r+b+c)(r+b+2c)} [r(r)(r+c) + r(b+c)(r+c) + b(r+c)(r+c) + b(r)(r+2c)].$$

Time-invariance fails. In the standard urn $P(R_n) = \frac{r}{r+b}$ for all n . Here, comparing $P(R_2)$ with $P(R_1)$:

$$P(R_2) - P(R_1) = \frac{r(r+b) + bc}{(r+b)(r+b+c)} - \frac{r}{r+b} = \frac{bc - rc}{(r+b)(r+b+c)} = \frac{c(b-r)}{(r+b)(r+b+c)}.$$

This is nonzero whenever $r \neq b$ (and $c > 0$), so $P(R_n)$ is **not** time-invariant. The opposite-color reinforcement pushes the red probability toward $\frac{1}{2}$: it rises when $b > r$ and falls when $r > b$.

Section C: Conditional Probability & Bayes

Question 7

Let the pardoned prisoner be uniform: $P(A) = P(B) = P(C) = \frac{1}{3}$. Let W_B be the event that the warden says “B will be executed.” The warden’s behavior:

- If A is pardoned, he may say B or C, each with prob $\frac{1}{2}$: $P(W_B | A) = \frac{1}{2}$.
- If B is pardoned, he cannot name B, so he says C: $P(W_B | B) = 0$.
- If C is pardoned, he is forced to say B: $P(W_B | C) = 1$.

By Bayes,

$$P(A | W_B) = \frac{P(W_B | A)P(A)}{\sum P(W_B | \cdot)P(\cdot)} = \frac{\frac{1}{2} \cdot \frac{1}{3}}{\frac{1}{2} \cdot \frac{1}{3} + 0 \cdot \frac{1}{3} + 1 \cdot \frac{1}{3}} = \frac{\frac{1}{6}}{\frac{1}{2}} = \frac{1}{3}.$$

So A is **incorrect**: his survival probability stays $\frac{1}{3}$, while $P(C | W_B) = \frac{2}{3}$.

Question 8

Alice wins exactly when the first Head occurs on an odd toss (tosses 1, 3, 5, ...). Each such case is mutually exclusive, requiring $2k - 2$ misses (prob $(1 - p)^{2k-2}$) then a Head:

$$P(A) = \sum_{k=1}^{\infty} (1 - p)^{2k-2} p = \frac{p}{1 - (1 - p)^2} = \frac{p}{2p - p^2} = \frac{1}{2 - p}.$$

For a fair game, set $P(A) = \frac{1}{2}$:

$$\frac{1}{2 - p} = \frac{1}{2} \Leftrightarrow 2 - p = 2 \Leftrightarrow p = 0.$$

Since $p = 0 \notin (0, 1)$, **no** bias makes the game fair. Going first always gives $P(A) = \frac{1}{2 - p} > \frac{1}{2}$ for $p \in (0, 1)$.

Question 9

No. Mutual independence requires the triple product *and* all three pairwise products. The single triple equation is not enough.

Counterexample. Let $\Omega = \{1, 2, \dots, 8\}$ be equally likely (each outcome $\frac{1}{8}$), and set

$$A = \{1, 2, 3, 4\}, \quad B = \{1, 2, 5, 6\}, \quad C = \{1, 5, 7, 8\}.$$

Then $P(A) = P(B) = P(C) = \frac{1}{2}$. Since $A \cap B \cap C = \{1\}$,

$$P(A \cap B \cap C) = \frac{1}{8} = P(A)P(B)P(C),$$

so the triple product holds. But $A \cap C = \{1\}$ gives

$$P(A \cap C) = \frac{1}{8} \neq \frac{1}{4} = P(A)P(C),$$

so A and C are not even pairwise independent. Mutual independence fails despite the triple product.

Section D: CDF Properties and Discontinuities

Question 10

A CDF F is nondecreasing with $0 \leq F \leq 1$. At a discontinuity x , the jump size is

$$J(x) = F(x) - F(x^-) = P(X = x) > 0.$$

For each $n \geq 1$, let $D_n = \{x : J(x) > \frac{1}{n}\}$. Since the jumps are disjoint mass and total mass is 1, D_n can contain at most n points (more than n jumps each exceeding $\frac{1}{n}$ would sum past 1). So each D_n is finite.

The set of all discontinuities is

$$D = \bigcup_{n=1}^{\infty} D_n,$$

a countable union of finite sets, hence at most countable.

Question 11

Enumerate the rationals in $[0, 1]$ as $q_1, q_2, q_3, \dots$ and assign mass $p_k = 2^{-k}$, so $\sum_k p_k = 1$. Define

$$F(x) = \sum_{k: q_k \leq x} 2^{-k}.$$

This is a valid CDF: nondecreasing, right-continuous, $F(x) \rightarrow 0$ as $x \rightarrow -\infty$ and $F(x) \rightarrow 1$ as $x \rightarrow \infty$. The distribution is discrete with support $\mathbb{Q} \cap [0, 1]$.

It is strictly increasing on $(0, 1)$: for any $a < b$ in $(0, 1)$, the interval $(a, b]$ contains some rational q_k , which carries mass $2^{-k} > 0$, so $F(b) - F(a) \geq 2^{-k} > 0$.

Question 12

The point is uniform on the unit disk (area π). For $0 \leq x \leq 1$, the event $X \leq x$ is the sub-disk of radius x (area πx^2):

$$F_X(x) = \frac{\pi x^2}{\pi} = x^2, \quad 0 \leq x \leq 1,$$

with $F_X(x) = 0$ for $x < 0$ and $F_X(x) = 1$ for $x > 1$. Differentiating,

$$f_X(x) = 2x, \quad 0 \leq x \leq 1.$$

Question 13

$X \sim \text{Unif}(-1, 1)$ and $Y = X^2$, so $Y \in [0, 1)$. For $0 \leq y < 1$,

$$F_Y(y) = P(X^2 \leq y) = P(-\sqrt{y} \leq X \leq \sqrt{y}) = \int_{-\sqrt{y}}^{\sqrt{y}} \frac{1}{2} dx = \sqrt{y}.$$

Differentiating,

$$f_Y(y) = \frac{1}{2\sqrt{y}}, \quad 0 < y < 1,$$

and 0 elsewhere. The density diverges as $y \rightarrow 0^+$ because squaring compresses values near zero.

Section E: PDFs, Quantiles, and Mixed Random Variables

Question 14

False. A PDF need only satisfy $f \geq 0$ and $\int f = 1$; it may be unbounded as long as the area stays finite.

Construction (unbounded at infinitely many points of $[0, 1]$). Place a shrinking integrable spike at each point $\frac{1}{n}$. Define

$$f(x) = \sum_{n=1}^{\infty} \frac{c_n}{2\sqrt{|x - \frac{1}{n}|}} \mathbf{1}(|x - \frac{1}{n}| < a_n),$$

where the windows a_n are small enough to be disjoint and we choose weights so each spike has area 2^{-n} . A single spike $\frac{c_n}{2\sqrt{|x - 1/n|}}$ over a window of half-width a_n contributes

$$\int \frac{c_n}{2\sqrt{|x - 1/n|}} dx = c_n \cdot 2\sqrt{a_n},$$

so setting $c_n = \frac{2^{-n}}{2\sqrt{a_n}}$ gives area 2^{-n} per spike. Then $f \geq 0$ and

$$\int_0^1 f = \sum_{n=1}^{\infty} 2^{-n} = 1,$$

while $f \rightarrow \infty$ at each $x = \frac{1}{n}$, i.e. at infinitely many points.

Question 15

Condition on X_2 and use independence:

$$P(X_1 = X_2) = \int_{-\infty}^{\infty} P(X_1 = x) dF(x).$$

Since F is continuous, X_1 has no atoms: $P(X_1 = x) = F(x) - F(x^-) = 0$ for every x . To see this from limits, take $x_n \downarrow x$; continuity gives $F(x_n) \rightarrow F(x)$, so

$$P(X_1 = x) = \lim_n P(x < X_1 \leq x_n) = \lim_n (F(x_n) - F(x)) = 0.$$

The integrand is identically 0, hence

$$P(X_1 = X_2) = \int 0 dF(x) = 0.$$

Question 16

The CDF of X has an atom of size p at 0, then continuous mass governed by G :

$$F_X(x) = \begin{cases} 0 & x < 0 \\ p + (1-p)G(x) & x \geq 0 \end{cases}$$

(using $G(0) = 0$). The q -th quantile is $\xi_q = \inf\{x : F_X(x) \geq q\}$.

- If $q \leq p$: the jump at 0 already reaches q , so $\xi_q = 0$.
- If $q > p$: solve $p + (1-p)G(x) = q$, giving

$$\xi_q = G^{-1}\left(\frac{q-p}{1-p}\right).$$

Combining,

$$\xi_q = \begin{cases} 0 & 0 < q \leq p \\ G^{-1}\left(\frac{q-p}{1-p}\right) & p < q < 1 \end{cases}$$